ASHLESHA M. POKALE

ELECTRONICS & TELECOMMUNICATION ENGINEERING

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PROFILE

Electronics and Telecommunication Engineering undergraduate with hands-on experience in semiconductor device characterization, NGSpice simulation, and Python-based data analysis. Skilled in analyzing diode I–V, MOS capacitor C–V, and MOSFET characteristics, with additional experience in FPGA-based digital design using Verilog HDL and VHDL. Interested in semiconductor devices, electronic characterization, and electronics R&D.

EDUCATION HISTORY

St. Vincent Pallotti College of Engineering and Technology, Nagpur 2023-2027
B. Tech Electronics and Telecommunications Engineering
CGPA: 7.70 / 10

SEMICONDUCTOR DEVICE SIMULATION & CHARACTERIZATION

Semiconductor Devices – IIT Bombay JUN 2026

- Performed experimental I–V characterization of diodes and LEDs and analyzed solar-cell characteristics under dark and illuminated conditions.
- Analyzed MOS capacitor C–V characteristics and NMOS transfer, drain, body-effect, and subthreshold behavior.

Electronic Devices Simulation (NGSpice) & Analysis – IIT Bombay JUN 2026

- Simulated diode and Schottky diode I–V characteristics and analyzed semiconductor device behavior.
- Simulated NMOS, PMOS, CMOS inverter, and subthreshold characteristics using NGSpice.
- Analyzed simulation data to examine device and circuit responses.

PROJECTS

Indigenous Railway Track Monitoring System (ITMS) Final Year Project – Ongoing

- Designed and developed an indigenous, contactless railway track monitoring prototype using a Radxa Cubie A7A single-board computer, 4K MIPI-CSI camera, and 6-axis IMU.
- Prepared and annotated a railway-track image dataset covering nine classes; trained and evaluated an AI-based defect detection model, achieving a reported mAP@0.5 of 0.795.
- Deployed the trained model for inference on the Radxa board, working on edge-AI execution for railway track inspection.
- Integrated camera and IMU hardware for visual monitoring and inertial data acquisition, and implemented a monitoring dashboard for system outputs and inspection-result visualization.

Digital Design and Verification Using Verilog HDL and VHDL

- Implemented digital logic circuits, including half adders, full adders, and multiplexers, using Verilog HDL.
- Designed and simulated a sequence detector using a finite-state machine (FSM) in VHDL with Xilinx ISE 14.7.

Fire Alarm System – Circuit Simulation and Hardware Implementation

- Designed and simulated a fire alarm circuit using Multisim.
- Implemented a hardware prototype to demonstrate sensor-based fire detection and alarm activation.

Smart Parking System (ESP32, IoT)

- Designed an IoT-based system using IR sensors and ESP32 to detect parking occupancy
- Displayed real-time slot availability on an LCD interface

INTERNSHIPS

Research Intern at Maharashtra Remote Sensing Application Centre (MRSAC), Nagpur JUN 2026

Duration: 1 Month

- Explored GIS, remote sensing, and IoT applications for precision agriculture.
- Developed an interactive geospatial dashboard for agricultural monitoring.
- Implemented simulated sensor data, dynamic charts, soil-moisture thresholds, and irrigation logic using HTML, CSS, JavaScript, and Chart.js.
- Visualized agricultural parameters and a conceptual satellite-to-ground data comparison.

Industrial Automation with IIoT Intern at V. R. Jamdar Siemens Center of Excellence, VNIT Nagpur JUN 2025

Duration: 1 Month

- Practiced PLC programming using LAD, FBD, and SFC.
- Used Siemens TIA Portal for PLC configuration and automation programming.
- Explored SCADA/HMI-based process monitoring, visualization, and control.
- Studied industrial instrumentation and sensor–actuator interfacing.

TECHNICAL SKILLS

- Programming: C, C++, Python, Embedded C
- Semiconductor Devices & Simulation: Semiconductor Device Characterization, NGSpice, SPICE Simulation, Diode I–V Analysis, MOS Capacitor C–V Analysis, MOSFET Characteristics
- Digital Design: Verilog HDL, VHDL, FPGA Design, Finite-State Machines (FSM), Xilinx ISE 14.7
- Embedded Systems: ESP32, PIC/STM32 Microcontrollers, Sensor Interfacing
- Tools: MATLAB, Multisim, Microwind, Proteus, Siemens TIA Portal
- Web & Visualization: HTML, CSS, JavaScript, Chart.js